

Tarea 3 Solver

Ejercicio 1. Escribir las siguientes ecuaciones diferenciales de orden superior en un sistema lineal usando notación matricial.

1. $\frac{d^4x}{dt^4} - 10\frac{d^3x}{dt^3} + 35\frac{d^2x}{dt^2} - 50\frac{dx}{dt} + 24x = 0$
2. $\frac{d^4x}{dt^4} - 4\frac{d^3x}{dt^3} + 8\frac{d^2x}{dt^2} - 8\frac{dx}{dt} + 4x = 9\sin(2t)$

Solución.

(1) Definimos $x_1 = x$, $x_2 = \frac{dx}{dt}$, $x_3 = \frac{d^2x}{dt^2}$, $x_4 = \frac{d^3x}{dt^3}$. Entonces, $\dot{x}_1 = x_2$, $\dot{x}_2 = x_3$, $\dot{x}_3 = x_4$, $\dot{x}_4 = \frac{d^4x}{dt^4}$. Luego, la ecuación diferencial

$$\frac{d^4x}{dt^4} = 10\frac{d^3x}{dt^3} - 35\frac{d^2x}{dt^2} + 50\frac{dx}{dt} - 24x$$

Queda expresada como

$$\dot{x}_4 = 10x_4 - 35x_3 + 50x_2 - 24x_1$$

En forma matricial,

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -24 & 50 & -35 & 10 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}$$

(2) Definimos $x_1 = x$, $x_2 = \frac{dx}{dt}$, $x_3 = \frac{d^2x}{dt^2}$, $x_4 = \frac{d^3x}{dt^3}$. Entonces, $\dot{x}_1 = x_2$, $\dot{x}_2 = x_3$, $\dot{x}_3 = x_4$, $\dot{x}_4 = \frac{d^4x}{dt^4}$. Luego, la ED

$$\frac{d^4x}{dt^4} = 4\frac{d^3x}{dt^3} - 8\frac{d^2x}{dt^2} + 8\frac{dx}{dt} - 4x + 9\sin(2t)$$

Queda expresada como

$$\dot{x}_4 = 4x_4 - 8x_3 + 8x_2 - 4x_1 + 9\sin(2t)$$

En forma matricial,

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -4 & 8 & -8 & 4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 0 \\ 9\sin(2t) \end{pmatrix}$$

Ejercicio 2. Los siguientes vectores son soluciones de un sistema lineal homogéneo $\dot{x}(t) = Ax(t)$. Determinar si forman un conjunto fundamental de soluciones en $\mathbb{R}$.

1. $x_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^t$, $x_2 = \begin{pmatrix} 2 \\ 6 \end{pmatrix} e^t + \begin{pmatrix} 8 \\ -8 \end{pmatrix} te^t$
2. $x_1 = \begin{pmatrix} 1 \\ 6 \\ -13 \end{pmatrix}$, $x_2 = \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix} e^{-4t}$, $x_3 = \begin{pmatrix} 2 \\ 3 \\ -2 \end{pmatrix} e^{3t}$

Solución.

(1) Consideremos la matriz fundamental de soluciones:

$$X(t) = \begin{pmatrix} e^t & 2e^t + 8te^t \\ -e^t & 6e^t - 8te^t \end{pmatrix}.$$

El Wronskiano es

$$W(t) = \begin{vmatrix} e^t & 2e^t + 8te^t \\ -e^t & 6e^t - 8te^t \end{vmatrix} = e^t(6e^t - 8te^t) - (-e^t)(2e^t + 8te^t)$$

Factorizando e^{2t} ,

$$W(t) = e^{2t}(6 - 8t + 2 + 8t) = 8e^{2t}$$

Como

$$W(t) = 8e^{2t} \neq 0 \quad \forall t \in \mathbb{R}$$

los vectores son linealmente independientes para todo t , por lo que forman un conjunto fundamental de soluciones en $\mathbb{R}$.

(2) Consideremos la matriz fundamental

$$X(t) = \begin{pmatrix} 1 & e^{-4t} & 2e^{3t} \\ 6 & -2e^{-4t} & 3e^{3t} \\ -13 & -e^{-4t} & -2e^{3t} \end{pmatrix}$$

El Wronskiano es

$$W(t) = e^{-4t}e^{3t} \begin{vmatrix} 1 & 1 & 2 \\ 6 & -2 & 3 \\ -13 & -1 & -2 \end{vmatrix}$$

Calculamos el determinante:

$$\begin{aligned} \begin{vmatrix} 1 & 1 & 2 \\ 6 & -2 & 3 \\ -13 & -1 & -2 \end{vmatrix} &= 1 \begin{vmatrix} -2 & 3 \\ -1 & -2 \end{vmatrix} - 1 \begin{vmatrix} 6 & 3 \\ -13 & -2 \end{vmatrix} + 2 \begin{vmatrix} 6 & -2 \\ -13 & -1 \end{vmatrix} \\ &= 7 - 27 + 2(-32) \\ &= -84 \end{aligned}$$

Así,

$$W(t) = -84e^{-t} \neq 0 \quad \forall t \in \mathbb{R}$$

Por lo tanto, los tres vectores son linealmente independientes y forman un conjunto fundamental de soluciones en $\mathbb{R}$.

D

Ejercicio 3. Para cada uno de los siguientes sistemas,

- Mediante un análisis cualitativo basado en la traza y el determinante de la matriz de coeficientes, determine el tipo de punto crítico del sistema, dibuje el retrato de fases correspondiente (de acuerdo al plano traza-determinante) y establezca la estabilidad del punto de equilibrio (estable, asintóticamente estable o inestable).
- Encuentre la solución general.
- Describa el comportamiento de la solución general cuando $t \rightarrow \infty$.
- Dibuje a mano varias trayectorias en el plano de fases; es decir, construya el retrato de fases del sistema (*Hint: examine la dirección de los vectores propios y el signo de los valores propios asociados*).
- Utilice un dispositivo de graficación apropiado (por ejemplo, Matlab o Python) para graficar con precisión varias trayectorias en el plano de fases.

$$1. \dot{x}(t) = \begin{pmatrix} 1 & 1 \\ 4 & -2 \end{pmatrix} x(t)$$

$$2. \dot{x}(t) = \begin{pmatrix} 5 & -1 \\ 3 & 1 \end{pmatrix} x(t)$$

$$3. \dot{x}(t) = \begin{pmatrix} 1 & -2 \\ 3 & -4 \end{pmatrix} x(t)$$

$$4. \dot{x}(t) = \begin{pmatrix} 1 & -5 \\ 1 & -3 \end{pmatrix} x(t)$$

$$5. \dot{x}(t) = \begin{pmatrix} 2 & -1 \\ 3 & -2 \end{pmatrix} x(t)$$

$$6. \dot{x}(t) = \begin{pmatrix} 4 & -2 \\ 8 & -4 \end{pmatrix} x(t)$$

$$7. \dot{x}(t) = \begin{pmatrix} 2 & -5 \\ 1 & -2 \end{pmatrix} x(t)$$

$$8. \dot{x}(t) = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} x(t)$$

$$9. \dot{x}(t) = \begin{pmatrix} -1 & -1 \\ 0 & -\frac{1}{4} \end{pmatrix} x(t)$$

$$10. \dot{x}(t) = \begin{pmatrix} -\frac{3}{2} & -\frac{1}{4} \\ -1 & -\frac{3}{2} \end{pmatrix} x(t)$$

Solución.

$$(1) \dot{x}(t) = \begin{pmatrix} 1 & 1 \\ 4 & -2 \end{pmatrix} x(t)$$

Análisis cualitativo. Un punto crítico, punto de equilibrio o punto fijo de un sistema dinámico

$$\dot{\mathbf{x}} = F(\mathbf{x})$$

es un punto $\mathbf{x}^* \in \mathbb{R}^n$ tal que

$$F(\mathbf{x}^*) = \mathbf{0}$$

En macroeconomía, es habitual llamar **estado estacionario** a un punto de equilibrio del sistema dinámico. En el caso particular de un sistema lineal homogéneo con coeficientes constantes,

$$\dot{\mathbf{x}}(t) = A\mathbf{x}(t),$$

los puntos de equilibrio satisfacen

$$A\mathbf{x}^* = \mathbf{0}$$

Si la matriz A es invertible (equivalentemente, $\det(A) \neq 0$), el único estado estacionario es el origen:

$$\mathbf{x}^* = \mathbf{0}$$

Para estudiar cualitativamente este equilibrio en dimensión 2×2 , clasificamos el punto crítico y el retrato de fases mediante el análisis del plano traza–determinante, el cual relaciona los signos de

$$T = \text{Tr}(A), \quad D = \det(A)$$

con la naturaleza y estabilidad del equilibrio (véase la Figura 1). A continuación se presenta esta clasificación.

Teorema 1 (Clasificación mediante el Plano Traza–Determinante). Sea $A \in \mathbb{R}^{2 \times 2}$ y sea $\Delta = \text{Tr}^2(A) - 4\det(A)$ el discriminante de A . Entonces:

- Si $\det(A) < 0$, el equilibrio es un punto silla.
Es siempre inestable, independientemente del valor de $\text{Tr}(A)$.
- Si $\det(A) > 0$ y $\Delta > 0$, el equilibrio es un nodo.
 - Si $\text{Tr}(A) < 0$, es un nodo estable (nodo atractor).
 - Si $\text{Tr}(A) > 0$, es un nodo inestable (nodo repulsor).

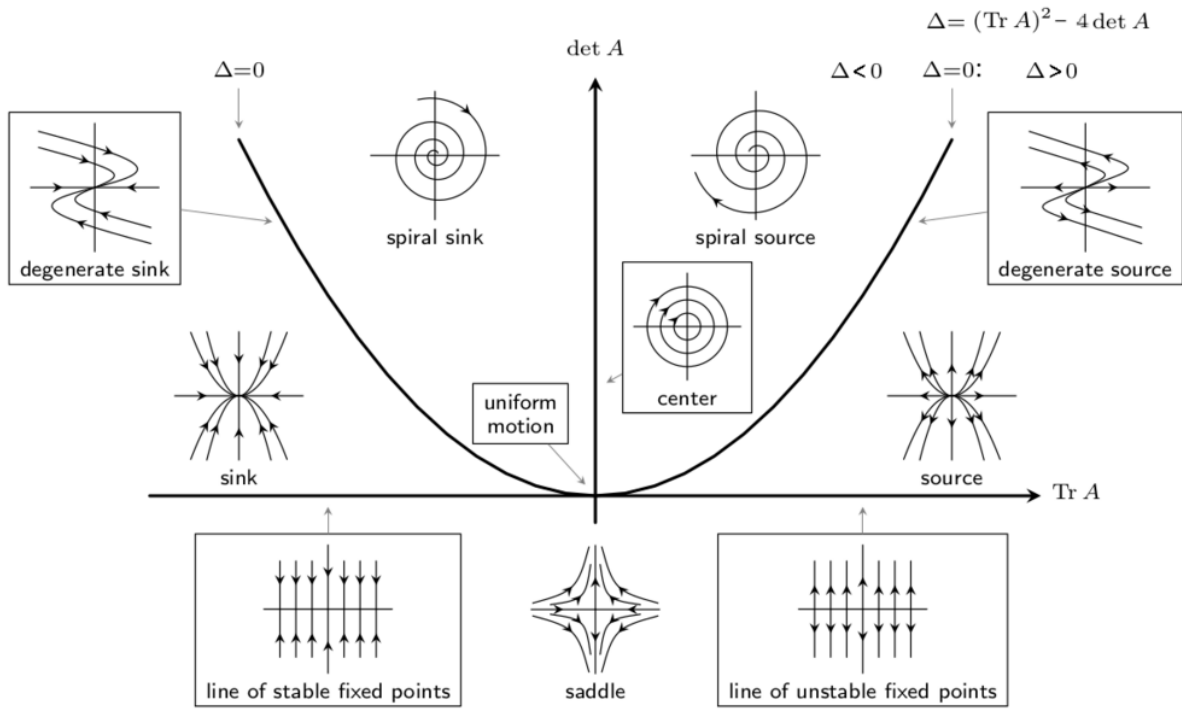


Figure 1: Clasificación de los retratos de fase en el Plano Traza-Determinante

- Si $\det(A) > 0$ y $\Delta < 0$, el equilibrio es un foco (o espiral).
 - Si $\text{Tr}(A) < 0$, es un foco estable (espiral atractora).
 - Si $\text{Tr}(A) > 0$, es un foco inestable (espiral repulsora).
- Si $\det(A) > 0$ y $\Delta = 0$, el equilibrio es un nodo degenerado.
 - Si $\text{Tr}(A) < 0$, es un nodo degenerado estable.
 - Si $\text{Tr}(A) > 0$, es un nodo degenerado inestable.
- Si $\det(A) > 0$, $\Delta < 0$, y $\text{Tr}(A) = 0$, el equilibrio es un centro. No es ni atractor ni repulsor; las trayectorias son órbitas cerradas que rodean al equilibrio.
- Si $\det(A) = 0$, el equilibrio consiste en una recta de puntos fijos.
 - Si $\text{Tr}(A) < 0$, la recta es estable.
 - Si $\text{Tr}(A) > 0$, la recta es inestable.

Nota: El plano Traza-Determinante proporciona una clasificación cualitativa de los sistemas lineales planos. Esta caracterización es específica de la dimensión 2×2 .

Por lo tanto, para este ejercicio tenemos que:

$$\text{Tr}(A) = \lambda_1 + \lambda_2 = -1,$$

$$\det(A) = \lambda_1 \lambda_2 = -6,$$

$$\Delta = \text{Tr}(A)^2 - 4 \det(A) = 25.$$



Figure 2: Retrato fase de un punto silla de acuerdo con el plano traza-determinante

Como $\det(A) < 0$, el punto crítico es un **punto silla**, independientemente del valor de $\text{Tr}(A)$.
Lo anterior se verifica también a partir de lo siguiente:

$$\lambda_{1,2} = \frac{\text{Tr}(A) \pm \sqrt{\text{Tr}^2(A) - 4\det(A)}}{2} = \frac{-1 \pm \sqrt{25}}{2} = \frac{-1 \pm 5}{2}$$

Claramente, $\lambda_1 < 0 < \lambda_2$ de modo que el punto crítico del sistema es un punto silla.

Todo punto silla es inestable en el sentido estricto, pero es estable en el sentido de silla, que significa que existe una única trayectoria asintóticamente estable, mientras todas las demás son inestables.

El retrato fase que deben dibujar en este punto es el de la Figura 2.

Solución general. Para la solución general, necesitamos calcular los valores y vectores propios de la matriz de coeficientes.

Ecuación característica, valores propios y vectores propios.

$$\begin{aligned} \det \begin{pmatrix} 1-\lambda & 1 \\ 4 & -2-\lambda \end{pmatrix} &= -(1-\lambda)(2+\lambda) - 4 \\ &= (\lambda-1)(2+\lambda) - 4 \\ &= 2\lambda + \lambda^2 - 2 - \lambda - 4 \\ &= \lambda^2 + \lambda - 6 = 0 \end{aligned}$$

Por lo tanto, la ecuación característica es

$$\mathcal{P}(\lambda) = \lambda^2 + \lambda - 6$$

Las raíces de la ecuación característica son

$$\lambda_{1,2} = \frac{-1 \pm \sqrt{1 - 4(-6)}}{2} = \frac{-1 \pm 5}{2}$$

Así, $\lambda_1 = 2$ y $\lambda_2 = -3$.

Esto ya lo habíamos encontrado en el inciso anterior, de una manera más rápida y útil, principalmente para casos en que la matriz contiene parámetros.

Para $\lambda_1 = 2$, el vector propio satisface

$$\begin{pmatrix} -1 & 1 \\ 4 & -4 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Es decir,

$$-v_1 + v_2 = 0 \quad \Rightarrow \quad v_1 = v_2$$

Un vector propio correspondiente es

$$V^{(1)} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Para $\lambda_2 = -3$, el vector propio satisface

$$\begin{pmatrix} 4 & 1 \\ 4 & 1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Es decir,

$$4v_1 + v_2 = 0 \quad \Rightarrow \quad 4v_1 = -v_2$$

Un vector propio correspondiente es

$$V^{(2)} = \begin{pmatrix} 1 \\ -4 \end{pmatrix}$$

Como los valores propios son distintos, la solución general es

$$x(t) = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{2t} + c_2 \begin{pmatrix} 1 \\ -4 \end{pmatrix} e^{-3t}$$

Dado que el sistema tiene dos valores propios distintos con signos distintos, el punto crítico $(0, 0)$ es un **punto silla**, como ya lo habíamos demostrado con el plano traza-determinante.

En particular, el sistema tiene una dirección propia inestable a lo largo de $V^{(1)} = (1, 1)^\top$. A menos que $c_1 = 0$, todas las soluciones divergen.

El subespacio propio generado por la dirección propia inestable se denomina **variedad inestable** (o brazo inestable, en el caso plano). El subespacio propio generado por la dirección propia estable se denomina **variedad estable** (o brazo estable, en el caso plano). La solución $x(t)$ converge al punto crítico (estado estacionario) sobre la variedad estable. La velocidad de convergencia está determinada por el valor propio estable. Sobre la variedad inestable, $\lim_{t \rightarrow \infty} x(t) = \infty$ o, equivalentemente, $\lim_{t \rightarrow -\infty} x(t) = 0$.

Comportamiento asintótico. Cuando $t \rightarrow \infty$, se tiene que

$$e^{2t} \rightarrow \infty \quad e^{-3t} \rightarrow 0,$$

lo cual implica que la solución general explota para cualquier condición inicial tal que $c_1 \neq 0$.

Retrato de fases a mano. Para dibujar el retrato de fases a mano, primero grafique los vectores propios asociados a los valores propios de A . Luego, dibuje la configuración tipo silla de acuerdo con dichas direcciones (ver Figura 1).

En este caso, como e^{2t} crece exponencialmente mientras que e^{-3t} decrece a cero, las trayectorias que se aproximan al estado estacionario lo hacen alineándose con la variedad estable (el subespacio propio asociado al valor propio negativo). Por el contrario, las trayectorias que se alejan del estado estacionario se alinean asintóticamente con la variedad inestable (el subespacio propio asociado al valor propio positivo).

La única trayectoria que converge al estado estacionario es aquella contenida en el subespacio generado por el vector propio estable; es decir, la variedad estable.

Retrato de fases con dispositivo de graficación. Véase la Figura 3.

$$(2) \dot{x}(t) = \begin{pmatrix} 5 & -1 \\ 3 & 1 \end{pmatrix} x(t)$$

Characteristic equation, eigenvalues and eigenvectors.

$$\begin{aligned} \det \begin{pmatrix} 5 - \lambda & -1 \\ 3 & 1 - \lambda \end{pmatrix} &= (5 - \lambda)(1 - \lambda) + 3 \\ &= 5 - 5\lambda - \lambda + \lambda^2 + 3 \\ &= \lambda^2 - 6\lambda + 8 = 0 \end{aligned}$$

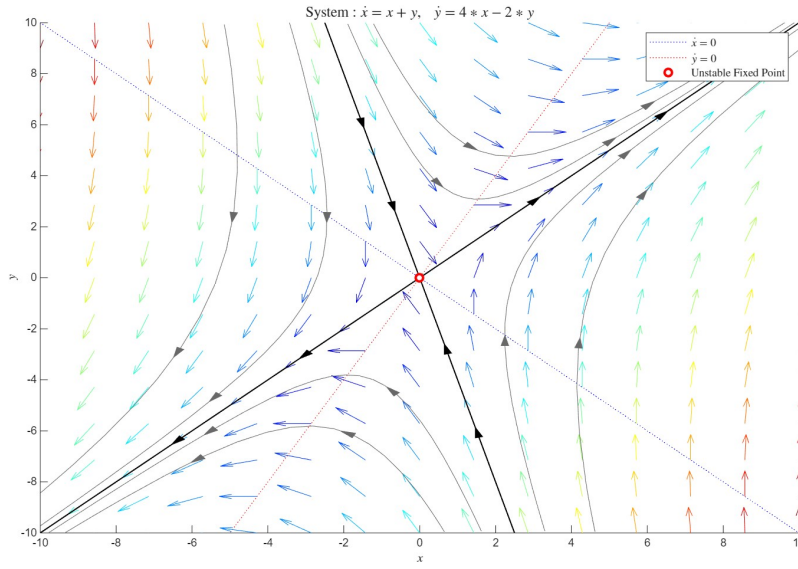


Figure 3: Figura generada usando Phase Portrait Plotter en Matlab.

Therefore, the characteristic equation is

$$\mathcal{P}(\lambda) = \lambda^2 - 6\lambda + 8$$

The roots of the characteristic equation are

$$\lambda_{1,2} = \frac{6 \pm \sqrt{36 - 32}}{2} = \frac{6 \pm 2}{2}$$

Hence,

$$\lambda_1 = 4, \quad \lambda_2 = 2$$

For $\lambda_1 = 4$, the eigenvector satisfies

$$\begin{pmatrix} 5 - 4 & -1 \\ 3 & 1 - 4 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

That is,

$$v_1 - v_2 = 0 \quad \Rightarrow \quad v_1 = v_2$$

A corresponding eigenvector is

$$V^{(1)} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

For $\lambda_2 = 2$, the eigenvector satisfies

$$\begin{pmatrix} 5 - 2 & -1 \\ 3 & 1 - 2 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

That is,

$$3v_1 - v_2 = 0 \quad \Rightarrow \quad v_2 = 3v_1$$

A corresponding eigenvector is

$$V^{(2)} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

Since the eigenvalues are distinct, the general solution is

$$x(t) = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{4t} + c_2 \begin{pmatrix} 1 \\ 3 \end{pmatrix} e^{2t}$$

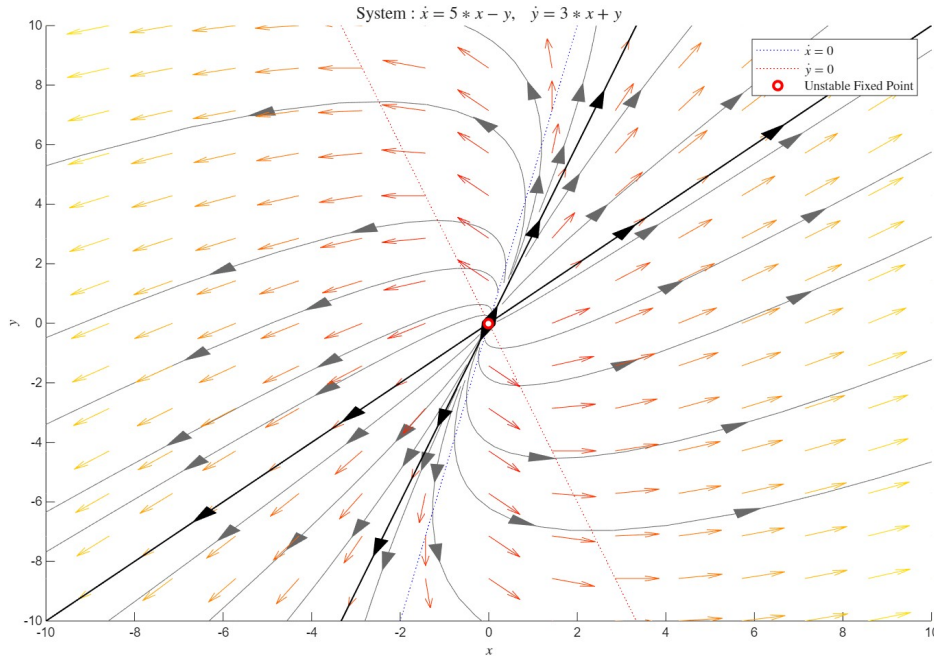


Figure 4: Figure generated using Phase Portrait Plotter in Matlab.

Critical point. Since both eigenvalues are real and strictly positive, the critical point $(0,0)$ is an **unstable node**.

All trajectories move away from the steady state along directions determined by the eigenvectors.

Asymptotic behavior. As $t \rightarrow \infty$, we have

$$e^{4t} \rightarrow \infty, \quad e^{2t} \rightarrow \infty$$

which implies that every nontrivial solution diverges.

Moreover, since e^{4t} grows faster than e^{2t} , the component of the solution associated with the eigenvalue $\lambda_1 = 4$ eventually dominates the dynamics. As trajectories move away from the steady state, they become asymptotically aligned with the eigenspace associated with the largest eigenvalue, namely $V^{(1)} = (1, 1)^\top$.

Phase portrait by hand. We classify the phase portrait using the trace-determinant plane:

$$\text{Tr}(A) = \lambda_1 + \lambda_2 = 6$$

$$\det(A) = \lambda_1 \lambda_2 = 8$$

$$\Delta = \text{Tr}(A)^2 - 4 \det(A) = 4$$

Since $\det(A) > 0$ and $\Delta > 0$, the steady state is a **node**. Since $\text{Tr}(A) > 0$, it is an **unstable node**.

To sketch the phase portrait by hand, first plot the eigenvectors and then draw representative node-like trajectories according to their directions.

The visible shape of the trajectories depends on the relative orientation of the eigenvectors. If the eigenvectors are nearly orthogonal, trajectories appear almost straight. If the eigenvectors are highly oblique, trajectories appear strongly curved. In all cases, trajectories become asymptotically aligned with the eigenspace associated with the eigenvalue that dominates the dynamics, namely the eigenvalue with the largest real part.

Phase portrait by graphing device. See Figure 4.

$$(3) \quad \dot{x}(t) = \begin{pmatrix} 1 & -2 \\ 3 & -4 \end{pmatrix} x(t)$$

Characteristic equation, eigenvalues and eigenvectors.

$$\begin{aligned}\det \begin{pmatrix} 1-\lambda & -2 \\ 3 & -4-\lambda \end{pmatrix} &= (1-\lambda)(-4-\lambda) + 6 \\ &= -4 - \lambda + 4\lambda + \lambda^2 + 6 \\ &= \lambda^2 + 3\lambda + 2 = 0\end{aligned}$$

Therefore, the characteristic equation is

$$\mathcal{P}(\lambda) = \lambda^2 + 3\lambda + 2$$

The roots of the characteristic equation are

$$\lambda_{1,2} = \frac{-3 \pm \sqrt{9-8}}{2} = \frac{-3 \pm 1}{2}$$

Hence,

$$\lambda_1 = -1, \quad \lambda_2 = -2$$

For $\lambda_1 = -1$, the eigenvector satisfies

$$\begin{pmatrix} 1+1 & -2 \\ 3 & -4+1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

That is,

$$2v_1 - 2v_2 = 0 \quad \Rightarrow \quad v_1 = v_2$$

A corresponding eigenvector is

$$V^{(1)} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

For $\lambda_2 = -2$, the eigenvector satisfies

$$\begin{pmatrix} 1+2 & -2 \\ 3 & -4+2 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

That is,

$$3v_1 - 2v_2 = 0 \quad \Rightarrow \quad v_2 = \frac{3}{2}v_1$$

A corresponding eigenvector is

$$V^{(2)} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

Since the eigenvalues are distinct, the general solution is

$$x(t) = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{-t} + c_2 \begin{pmatrix} 2 \\ 3 \end{pmatrix} e^{-2t}$$

Critical point. Since both eigenvalues are real and strictly negative, the critical point $(0,0)$ is a **stable node**.

Asymptotic behavior. As $t \rightarrow \infty$, we have

$$e^{-t} \rightarrow 0, \quad e^{-2t} \rightarrow 0$$

which implies that every solution converges to the steady state.

Moreover, since e^{-2t} decays faster than e^{-t} , the component of the solution associated with the eigenvalue $\lambda_1 = -1$ eventually dominates the dynamics. As trajectories approach the steady state, they become asymptotically aligned with the eigenspace associated with the eigenvalue with largest real part (the slow decay direction), namely $V^{(1)} = (1,1)^\top$.

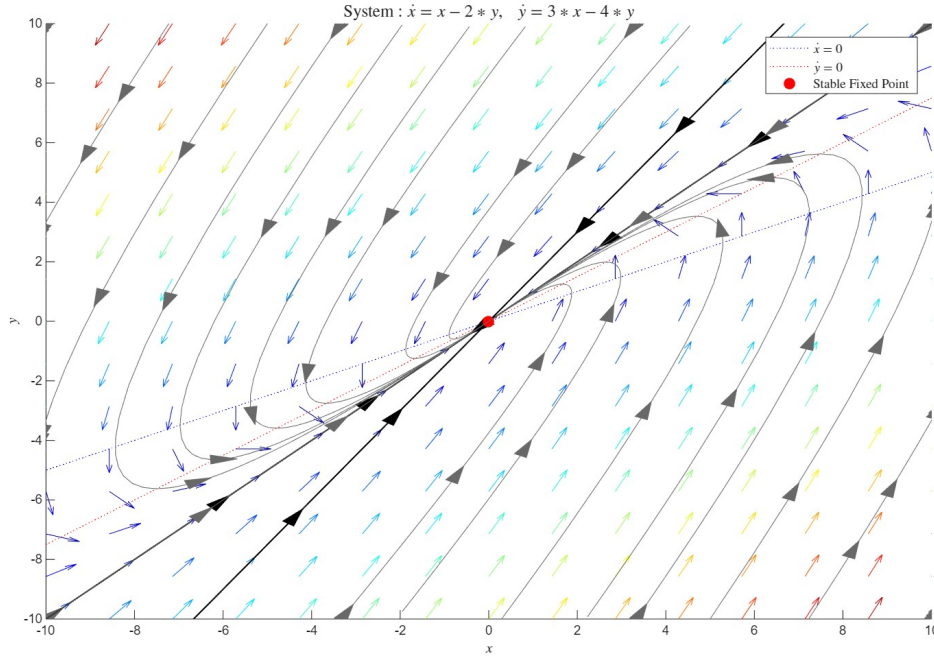


Figure 5: Figure generated using Phase Portrait Plotter in Matlab.

Phase portrait by hand. We classify the phase portrait using the trace–determinant plane:

$$\text{Tr}(A) = \lambda_1 + \lambda_2 = -3$$

$$\det(A) = \lambda_1 \lambda_2 = 2$$

$$\Delta = \text{Tr}(A)^2 - 4 \det(A) = 1$$

Since $\det(A) > 0$ and $\Delta > 0$, the steady state is a **node**. Since $\text{Tr}(A) < 0$, it is an **unstable node**.

To sketch the phase portrait by hand, first plot the eigenvectors and then draw representative node-like trajectories according to their directions.

The visible shape of the trajectories depends on the relative orientation of the eigenvectors. If the eigenvectors are nearly orthogonal, trajectories appear almost straight. If the eigenvectors are highly oblique, trajectories appear strongly curved. In all cases, trajectories become asymptotically aligned with the eigenspace associated with the eigenvalue that dominates the dynamics, namely the eigenvalue with the largest real part (the less negative eigenvalue, for the attractor case).

Phase portrait by graphing device. See Figure 5.

$$(4) \dot{x}(t) = \begin{pmatrix} 1 & -5 \\ 1 & -3 \end{pmatrix} x(t)$$

Characteristic equation, eigenvalues and eigenvectors.

$$\begin{aligned} \det \begin{pmatrix} 1 - \lambda & -5 \\ 1 & -3 - \lambda \end{pmatrix} &= (1 - \lambda)(-3 - \lambda) + 5 \\ &= \lambda^2 + 2\lambda + 2 = 0 \end{aligned}$$

Therefore, the characteristic equation is

$$\mathcal{P}(\lambda) = \lambda^2 + 2\lambda + 2$$

The roots of the characteristic equation are

$$\lambda_{1,2} = \frac{-2 \pm \sqrt{4-8}}{2} = -1 \pm i$$

Hence, the system has a pair of complex conjugate eigenvalues

$$\lambda_{1,2} = -1 \pm i$$

Consider $\lambda = -1 + i$. The associated eigenvector satisfies

$$\begin{pmatrix} 2-i & -5 \\ 1 & -2-i \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

From the second row,

$$v_1 - (2+i)v_2 = 0 \Rightarrow v_1 = (2+i)v_2$$

Taking $v_2 = 1$,

$$v = \begin{pmatrix} 2+i \\ 1 \end{pmatrix} = U + iV$$

with

$$U = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad V = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

Real general solution (Theorem 3.48).

Since $\lambda = \alpha + i\beta = -1 + i$, with eigenvector $v = U + iV$, the real solutions are

$$W_1(t) = e^{-t} (U \cos t - V \sin t)$$

$$W_2(t) = e^{-t} (U \sin t + V \cos t)$$

Hence, the general real solution is

$$x(t) = c_1 W_1(t) + c_2 W_2(t)$$

that is,

$$x(t) = c_1 e^{-t} \left[\begin{pmatrix} 2 \\ 1 \end{pmatrix} \cos t - \begin{pmatrix} 1 \\ 0 \end{pmatrix} \sin t \right] + c_2 e^{-t} \left[\begin{pmatrix} 2 \\ 1 \end{pmatrix} \sin t + \begin{pmatrix} 1 \\ 0 \end{pmatrix} \cos t \right]$$

Asymptotic behavior. As $t \rightarrow \infty$, we have

$$e^{-t} \rightarrow 0$$

while the trigonometric terms remain bounded. Therefore, all trajectories spiral toward the steady state.

Critical point. Since the real part of the eigenvalues is negative and the eigenvalues are complex, the critical point $(0, 0)$ is a **stable focus**.

Phase portrait by hand. We classify the phase portrait using the trace-determinant plane:

$$\text{Tr}(A) = \lambda_1 + \lambda_2 = -2$$

$$\det(A) = \lambda_1 \lambda_2 = 2$$

$$\Delta = \text{Tr}(A)^2 - 4 \det(A) = -4$$

Since $\det(A) > 0$ and $\Delta < 0$, the steady state is a **focus**. Since $\text{Tr}(A) < 0$, it is an **stable focus**.

To sketch the phase portrait by hand, just plot some spirals that converge to the fixed point.

Phase portrait by graphing device. See Figure 6.

$$(5) \dot{x}(t) = \begin{pmatrix} 2 & -1 \\ 3 & -2 \end{pmatrix} x(t)$$

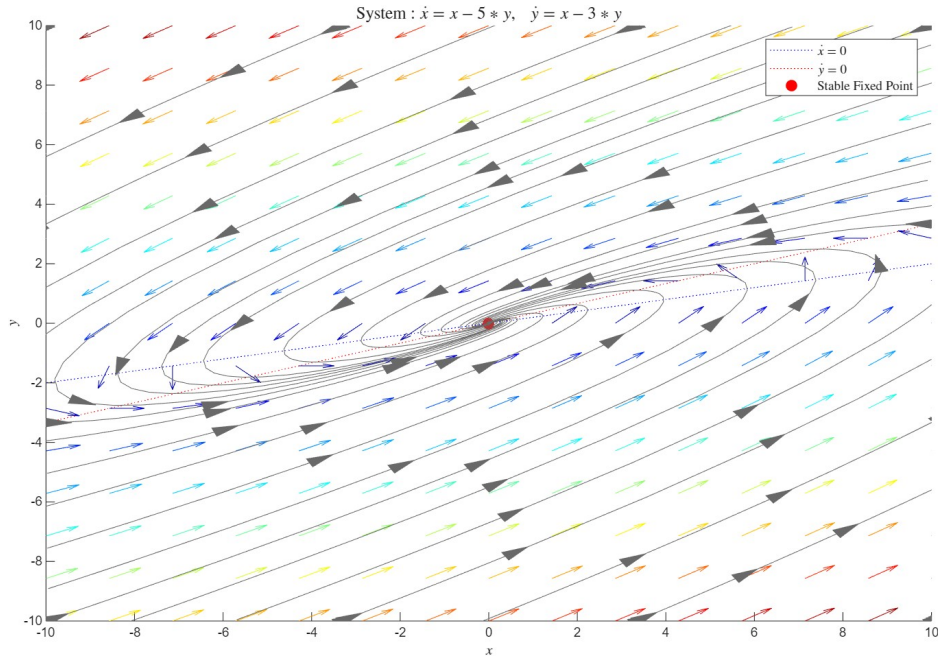


Figure 6: Figure generated using Phase Portrait Plotter in Matlab.

Characteristic equation, eigenvalues and eigenvectors.

$$\det \begin{pmatrix} 2 - \lambda & -1 \\ 3 & -2 - \lambda \end{pmatrix} = (2 - \lambda)(-2 - \lambda) + 3$$

$$= \lambda^2 - 1 = 0$$

Therefore, the characteristic equation is

$$\mathcal{P}(\lambda) = \lambda^2 - 1$$

The roots of the characteristic equation are

$$\lambda_1 = 1, \quad \lambda_2 = -1$$

For $\lambda_1 = 1$, the eigenvector satisfies

$$\begin{pmatrix} 2 - 1 & -1 \\ 3 & -2 - 1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

That is,

$$v_1 - v_2 = 0 \quad \Rightarrow \quad v_1 = v_2$$

A corresponding eigenvector is

$$V^{(1)} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

For $\lambda_2 = -1$, the eigenvector satisfies

$$\begin{pmatrix} 2 + 1 & -1 \\ 3 & -2 + 1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

That is,

$$3v_1 - v_2 = 0 \quad \Rightarrow \quad v_2 = 3v_1$$

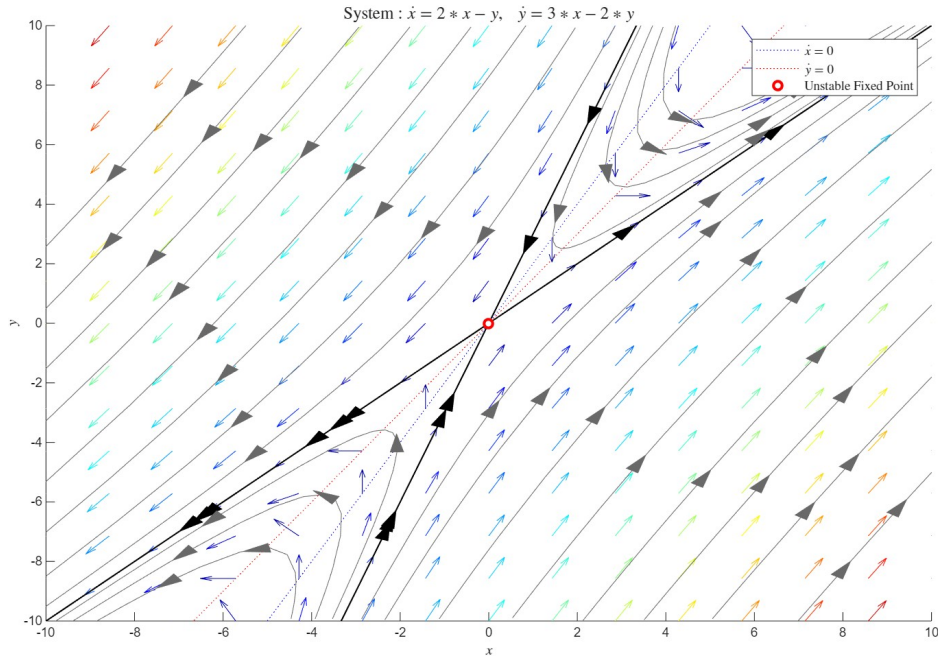


Figure 7: Figure generated using Phase Portrait Plotter in Matlab.

A corresponding eigenvector is

$$V^{(2)} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

Since the eigenvalues have opposite signs, the general solution is

$$x(t) = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^t + c_2 \begin{pmatrix} 1 \\ 3 \end{pmatrix} e^{-t}$$

Critical point. Since the eigenvalues are real with opposite signs, the critical point $(0,0)$ is a **saddle point**.

Asymptotic behavior. As $t \rightarrow \infty$, we have

$$e^t \rightarrow \infty, \quad e^{-t} \rightarrow 0$$

which implies that any solution with $c_1 \neq 0$ diverges.

Moreover, since e^t grows while e^{-t} decays, trajectories that move away from the steady state become asymptotically aligned with the unstable manifold, namely the space spanned by $V^{(1)} = (1, 1)^\top$.

Phase portrait by hand. We classify the phase portrait using the trace-determinant plane:

$$\text{Tr}(A) = \lambda_1 + \lambda_2 = 0$$

$$\det(A) = \lambda_1 \lambda_2 = -1$$

$$\Delta = \text{Tr}(A)^2 - 4 \det(A) = 4$$

Since $\det(A) < 0$, the steady state is a **saddle point**.

To sketch the phase portrait by hand, first plot the eigenvectors and then draw representative saddle point trajectories according to their directions.

Phase portrait by graphing device. See Figure 7.

$$(6) \dot{x}(t) = \begin{pmatrix} 4 & -2 \\ 8 & -4 \end{pmatrix} x(t)$$

Characteristic equation, eigenvalues and eigenvectors.

$$\begin{aligned}\det \begin{pmatrix} 4-\lambda & -2 \\ 8 & -4-\lambda \end{pmatrix} &= (4-\lambda)(-4-\lambda) + 16 \\ &= -16 - 4\lambda + 4\lambda + \lambda^2 \\ &= \lambda^2 = 0\end{aligned}$$

Hence, the characteristic equation is

$$\mathcal{P}(\lambda) = \lambda^2$$

and the system has a repeated eigenvalue

$$\lambda_1 = \lambda_2 = 0$$

To find the eigenvectors, solve

$$\begin{pmatrix} 4 & -2 \\ 8 & -4 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

From the first row,

$$4v_1 - 2v_2 = 0 \quad \Rightarrow \quad v_2 = 2v_1$$

A corresponding eigenvector is

$$V = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

Hence, one solution is

$$x^{(1)}(t) = e^{\lambda t} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

which is a constant vector.

Since there is only one linearly independent eigenvector, the matrix is defective. In order to generate a second linearly independent solution, we must search for a generalized eigenvector.

A generalized eigenvector W satisfies

$$(A - \lambda I)W = V$$

where V is the eigenvector associated with λ .

Hence, we need to solve the following system of equations

$$\begin{pmatrix} 4-\lambda & -2 \\ 8 & -4-\lambda \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

Since $\lambda = 0$, we obtain

$$\begin{pmatrix} 4 & -2 \\ 8 & -4 \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

From the first row, we get

$$w_1 = \frac{1}{4} + \frac{1}{2}w_2$$

and taking $w_2 = 0$ yields a generalized eigenvector

$$W = \begin{pmatrix} \frac{1}{4} \\ 0 \end{pmatrix}$$

Hence, the general solution is

$$x(t) = c_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + c_2 \left[\begin{pmatrix} 1 \\ 2 \end{pmatrix} t + \begin{pmatrix} \frac{1}{4} \\ 0 \end{pmatrix} \right]$$

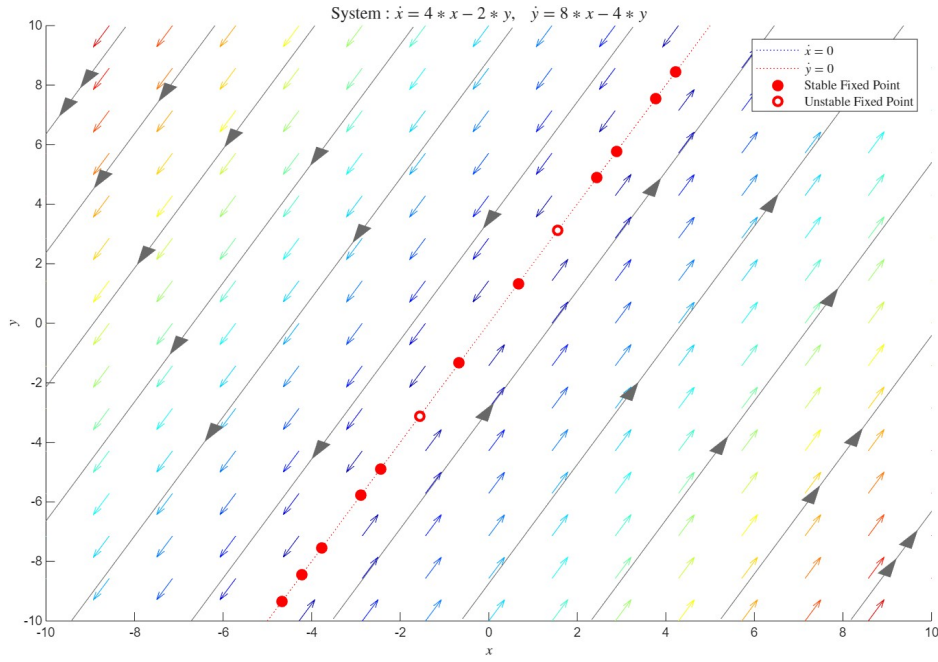


Figure 8: Figure generated using Phase Portrait Plotter in Matlab. All fixed points are stable. Phase Portrait Plotter couldn't identify the stability class.

Critical point. Since the only eigenvalue is $\lambda = 0$ and the matrix is defective, the critical point $(0,0)$ is a **degenerate (improper) node**.

Asymptotic behavior. Trajectories grow linearly in time unless $c_2 = 0$. Hence, for generic initial conditions,

$$\|x(t)\| \rightarrow \infty \quad \text{as} \quad t \rightarrow \infty,$$

and trajectories are aligned with the eigenspace spanned by $V = (1, 2)^\top$.

Phase portrait by hand. We classify the phase portrait using the trace-determinant plane:

$$\text{Tr}(A) = \lambda_1 + \lambda_2 = 0$$

$$\det(A) = \lambda_1 \lambda_2 = 0$$

$$\Delta = \text{Tr}(A)^2 - 4 \det(A) = 0$$

Since $\det(A) = \text{Tr}(A) = 0$, the system has a repeated eigenvalue $\lambda = 0$ and is defective. As a consequence, the eigenspace is one-dimensional and forms a **line of fixed points**.

To sketch the phase portrait by hand, first plot the eigenspace, which represents the line of steady states. Then draw straight lines parallel to this eigenspace. These lines represent the trajectories of the system, whose direction depends on the sign of the constant c_2 in the general solution.

Phase portrait by graphing device. See Figure 8.

$$(7) \quad \dot{x}(t) = \begin{pmatrix} 2 & -5 \\ 1 & -2 \end{pmatrix} x(t)$$

Characteristic equation, eigenvalues and eigenvectors.

$$\begin{aligned}\det \begin{pmatrix} 2-\lambda & -5 \\ 1 & -2-\lambda \end{pmatrix} &= (2-\lambda)(-2-\lambda) + 5 \\ &= -4 - 2\lambda + 2\lambda + \lambda^2 + 5 \\ &= \lambda^2 + 1 = 0\end{aligned}$$

Hence, the characteristic equation is

$$\mathcal{P}(\lambda) = \lambda^2 + 1$$

and the eigenvalues are

$$\lambda_{1,2} = \pm i$$

Consider $\lambda = i$. The eigenvector satisfies

$$\begin{pmatrix} 2-i & -5 \\ 1 & -2-i \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

From the second row,

$$v_1 - (2+i)v_2 = 0 \quad \Rightarrow \quad v_1 = (2+i)v_2$$

Taking $v_2 = 1$,

$$v = \begin{pmatrix} 2+i \\ 1 \end{pmatrix} = U + iV$$

where

$$U = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad V = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

Real general solution.

Since $\lambda = \alpha + i\beta = 0 + i$, by Theorem 3.48 the real solutions are

$$\begin{aligned}W_1(t) &= U \cos t - V \sin t \\ W_2(t) &= U \sin t + V \cos t\end{aligned}$$

Hence, the general solution is

$$x(t) = c_1 W_1(t) + c_2 W_2(t)$$

Critical point. Since the eigenvalues are purely imaginary and their real part is zero, the critical point $(0, 0)$ is a **center**. The steady state is Lyapunov stable, but not asymptotically stable.

Asymptotic behavior. Because there is no exponential term, all trajectories remain bounded for all t and move along closed orbits around the steady state.

Phase portrait by hand. We classify the phase portrait using the trace-determinant plane:

$$\text{Tr}(A) = \lambda_1 + \lambda_2 = 0$$

$$\det(A) = \lambda_1 \lambda_2 = 1$$

$$\Delta = \text{Tr}(A)^2 - 4 \det(A) = -4$$

Since $\det(A) > 0$, $\Delta < 0$, and $\text{Tr}(A) = 0$, the equilibrium point is a **center**.

To sketch the phase portrait by hand, draw closed trajectories around the steady state. These trajectories are ellipses whose orientation depends on the eigenvectors of A .

Phase portrait by graphing device. See Figure 9.

$$(8) \quad \dot{x}(t) = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} x(t)$$

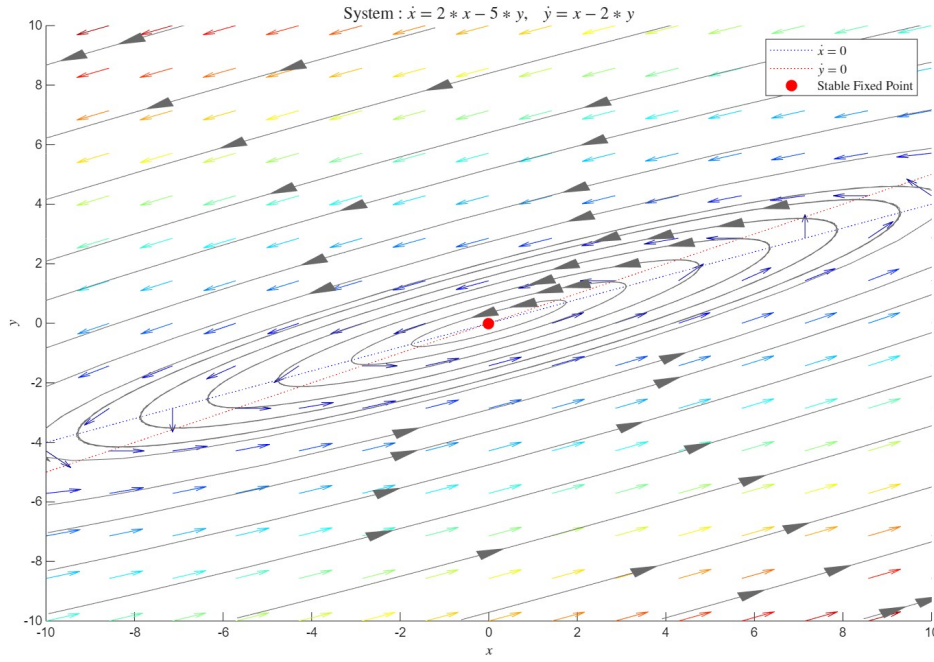


Figure 9: Figure generated using Phase Portrait Plotter in Matlab.

Characteristic equation, eigenvalues and eigenvectors.

$$\det \begin{pmatrix} -1 - \lambda & 0 \\ 0 & -1 - \lambda \end{pmatrix} = (1 + \lambda)^2 = 0$$

Hence, the characteristic equation is

$$\mathcal{P}(\lambda) = (1 + \lambda)^2$$

and the system has a repeated eigenvalue

$$\lambda_1 = \lambda_2 = -1$$

To find the eigenvectors, solve

$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

which imposes no restriction on v_1 and v_2 . That is, every vector in $\mathbb{R}^2$ is an eigenvector associated with $\lambda = -1$. Therefore, we can take two linearly independent vectors in $\mathbb{R}^2$ to construct two linearly independent solutions. In particular, take

$$V^{(1)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad V^{(2)} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Hence, the general solution is

$$x(t) = e^{-t} \left[c_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right] = e^{-t} x(0)$$

where we can see that all trajectories are simply some initial condition vector $x(0) = (c_1, c_2)^\top$ scaled by e^{-t} .

Critical point. Since there are two linearly independent eigenvectors, the critical point is an asymptotically stable proper node (stable star).

Asymptotic behavior. From the general solution we see that

$$\lim_{t \rightarrow \infty} x(t) = 0$$

All trajectories converge to the steady state along straight lines determined by the initial condition.

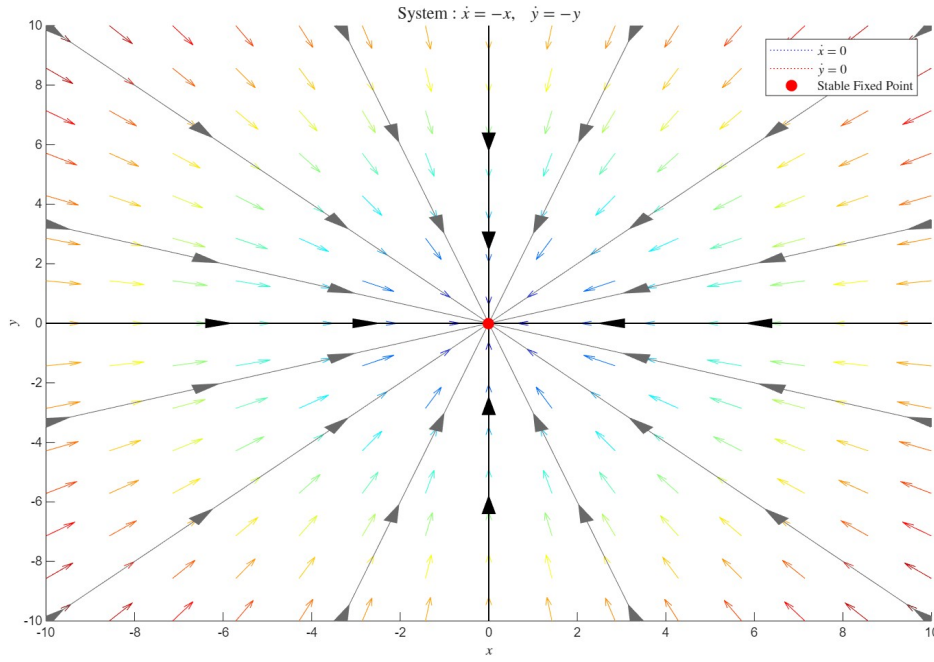


Figure 10: Figure generated using Phase Portrait Plotter in Matlab.

Phase portrait by hand. We classify the phase portrait using the trace–determinant plane:

$$\text{Tr}(A) = \lambda_1 + \lambda_2 = -2$$

$$\det(A) = \lambda_1 \lambda_2 = 1$$

$$\Delta = \text{Tr}(A)^2 - 4 \det(A) = 0$$

Since $\det(A) > 0$ and $\Delta = 0$ and there are two linearly independent eigenvectors, the node is a **star node**. Since $\text{Tr}(A) < 0$, it is stable.

To sketch the phase portrait by hand, plot the eigenvectors. All trajectories are straight lines converging to the steady state along directions determined by the initial condition.

Phase portrait by graphing device. All trajectories are straight lines pointing toward the origin, with uniform exponential decay in every direction. See Figure 10.

$$(9) \quad \dot{x}(t) = \begin{pmatrix} -1 & -1 \\ 0 & -\frac{1}{4} \end{pmatrix} x(t)$$

Characteristic equation, eigenvalues and eigenvectors.

$$\det \begin{pmatrix} -1 - \lambda & -1 \\ 0 & -\frac{1}{4} - \lambda \end{pmatrix} = (1 + \lambda)(\frac{1}{4} + \lambda) = 0$$

Clearly, the eigenvalues are

$$\lambda_1 = -1, \quad \lambda_2 = -\frac{1}{4}$$

For $\lambda_1 = -1$, solve $(A - \lambda_1 I)v = 0$:

$$\begin{pmatrix} 0 & -1 \\ 0 & \frac{3}{4} \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow v_2 = 0$$

Thus an eigenvector is

$$V^{(1)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

For $\lambda_2 = -\frac{1}{4}$, solve $(A - \lambda_2 I)v = 0$:

$$\begin{pmatrix} -\frac{3}{4} & -1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow v_2 = -\frac{3}{4}v_1$$

Thus an eigenvector is

$$V^{(2)} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$$

Since the eigenvalues are real, distinct, and negative, the general solution is

$$x(t) = c_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{-t} + c_2 \begin{pmatrix} 4 \\ -3 \end{pmatrix} e^{-t/4}$$

Critical point. Since both eigenvalues are real, distinct and strictly negative, the critical point $(0,0)$ is a **stable node**.

Asymptotic behavior. As $t \rightarrow \infty$,

$$e^{-t} \rightarrow 0 \quad e^{-t/4} \rightarrow 0$$

so every solution converges to the steady state.

Moreover, since e^{-t} decays faster than $e^{-t/4}$, the component associated with $\lambda_2 = -\frac{1}{4}$ (the slow decay direction) eventually dominates. As trajectories approach the steady state, they become asymptotically aligned with the eigenspace spanned by

$$V^{(2)} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$$

Phase portrait by hand. We classify the phase portrait using the trace-determinant plane:

$$\text{Tr}(A) = \lambda_1 + \lambda_2 = -\frac{5}{4}$$

$$\det(A) = \lambda_1 \lambda_2 = \frac{1}{4}$$

$$\Delta = \text{Tr}(A)^2 - 4\det(A) = \frac{9}{16}$$

Since $\det(A) > 0$ and $\Delta > 0$, the steady state is a **node**. Since $\text{Tr}(A) < 0$, it is an **stable node**.

To sketch the phase portrait by hand, first plot the eigenvectors and then draw representative node-like trajectories according to their directions.

The visible shape of the trajectories depends on the relative orientation of the eigenvectors. If the eigenvectors are nearly orthogonal, trajectories appear almost straight. If the eigenvectors are highly oblique, trajectories appear strongly curved. In all cases, trajectories become asymptotically aligned with the eigenspace associated with the eigenvalue that dominates the dynamics, namely the eigenvalue with the largest real part (the less negative eigenvalue, for the attractor case).

Phase portrait by graphing device. Figure 11.

$$(10) \quad \dot{x}(t) = \begin{pmatrix} -\frac{3}{2} & -\frac{1}{4} \\ -1 & -\frac{3}{2} \end{pmatrix} x(t)$$

Characteristic equation, eigenvalues and eigenvectors.

$$\begin{aligned} \det \begin{pmatrix} -\frac{3}{2} - \lambda & -\frac{1}{4} \\ -1 & -\frac{3}{2} - \lambda \end{pmatrix} &= \left(\frac{3}{2} + \lambda\right)^2 - \frac{1}{4} \\ &= \frac{9}{4} + 3\lambda + \lambda^2 - \frac{1}{4} \\ &= \lambda^2 + 3\lambda + 2 = 0 \end{aligned}$$

The roots of the characteristic equation are

$$\lambda_{1,2} = \frac{-3 \pm \sqrt{9-8}}{2} = \frac{-3 \pm 1}{2}$$

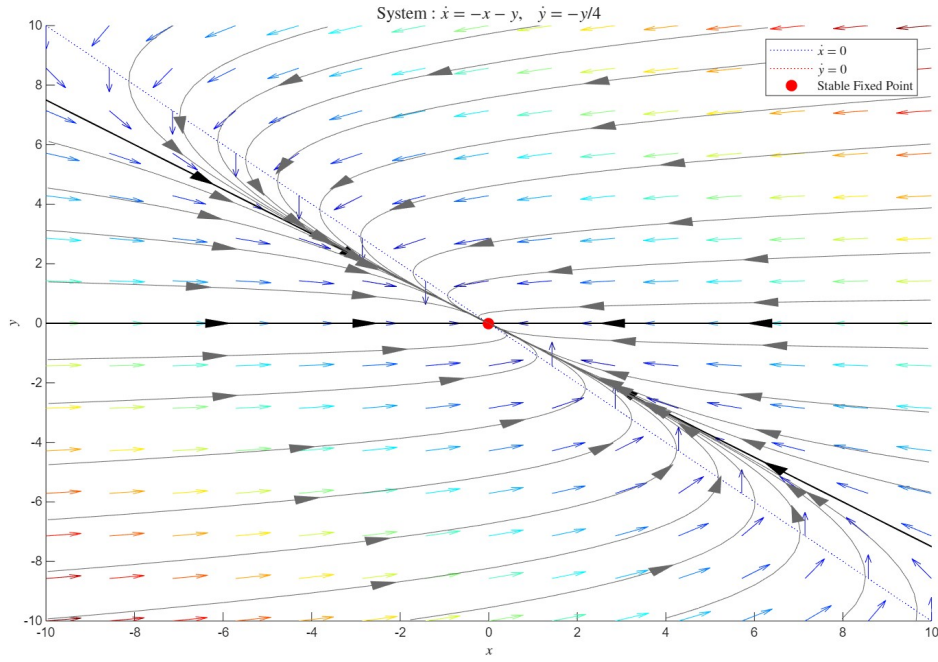


Figure 11: Figure generated using Phase Portrait Plotter in Matlab.

Hence,

$$\lambda_1 = -1, \quad \lambda_2 = -2$$

For $\lambda_1 = -1$, solve $(A - \lambda_1 I)v = 0$:

$$\begin{pmatrix} -\frac{1}{2} & -\frac{1}{4} \\ -1 & -\frac{3}{2} \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow 2v_1 = -v_2$$

Thus an eigenvector is

$$V^{(1)} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

For $\lambda_2 = -2$, solve $(A - \lambda_2 I)v = 0$:

$$\begin{pmatrix} \frac{1}{2} & -\frac{1}{4} \\ -1 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow 2v_1 = v_2$$

Thus an eigenvector is

$$V^{(2)} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

Since the eigenvalues are real, distinct, and negative, the general solution is

$$x(t) = c_1 \begin{pmatrix} -1 \\ 2 \end{pmatrix} e^{-t} + c_2 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{-2t}$$

Critical point. Because both eigenvalues are negative and distinct, the critical point $(0,0)$ is a **stable node**.

Asymptotic behavior. As $t \rightarrow \infty$,

$$e^{-t} \rightarrow 0, \quad e^{-2t} \rightarrow 0$$

so every solution converges to the steady state. Moreover, since e^{-2t} decays faster than e^{-t} , trajectories become asymptotically aligned with the eigenspace associated with $\lambda_1 = -1$, namely $V^{(1)} = (-1, 2)^\top$.

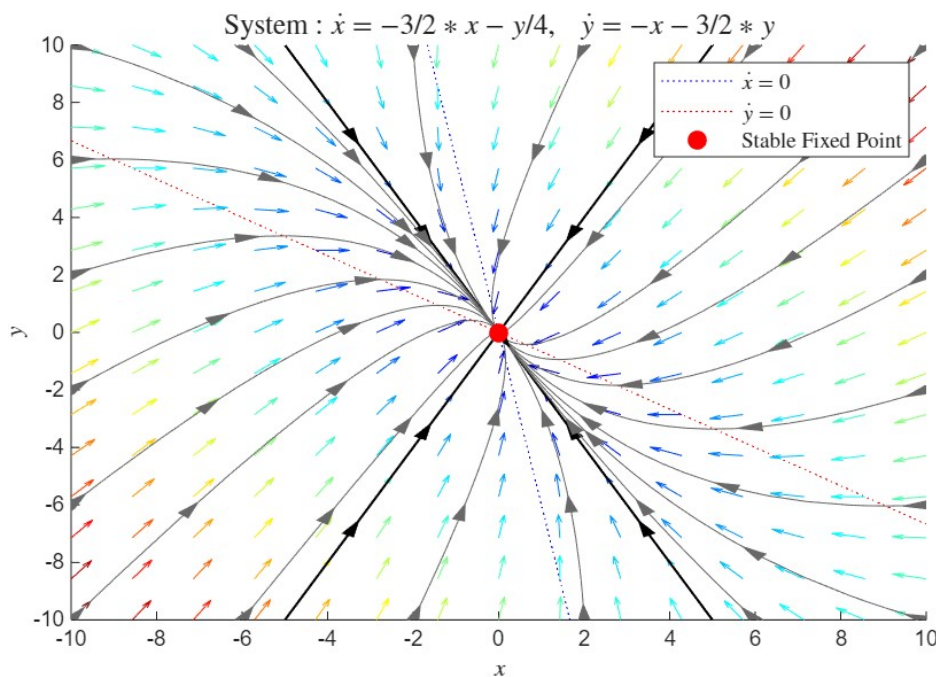


Figure 12: Figure generated using Phase Portrait Plotter in Matlab.

Phase portrait by hand. We classify the phase portrait using the trace-determinant plane:

$$\text{Tr}(A) = \lambda_1 + \lambda_2 = -3$$

$$\det(A) = \lambda_1 \lambda_2 = 2$$

$$\Delta = \text{Tr}(A)^2 - 4 \det(A) = 1$$

Since $\det(A) > 0$ and $\Delta > 0$, the steady state is a **node**. Since $\text{Tr}(A) < 0$, it is an **stable node**.

To sketch the phase portrait by hand, first plot the eigenvectors and then draw representative node-like trajectories according to their directions.

The visible shape of the trajectories depends on the relative orientation of the eigenvectors. If the eigenvectors are nearly orthogonal, trajectories appear almost straight. If the eigenvectors are highly oblique, trajectories appear strongly curved. In all cases, trajectories become asymptotically aligned with the eigenspace associated with the eigenvalue that dominates the dynamics, namely the eigenvalue with the largest real part (the less negative eigenvalue, for the attractor case).

Phase portrait by graphing device. Figure 12.

Ejercicio 4. Resolver los siguientes sistemas lineales homogéneos con valores iniciales.

$$1. Y' = \begin{pmatrix} 1 & -3 \\ -2 & 2 \end{pmatrix} Y, \quad Y(0) = \begin{pmatrix} 0 \\ 5 \end{pmatrix} \quad 2. Y' = \begin{pmatrix} 3 & 1 & -2 \\ -1 & 2 & 1 \\ 4 & 1 & -3 \end{pmatrix} Y, \quad Y(0) = \begin{pmatrix} 1 \\ 4 \\ 7 \end{pmatrix}$$

Solución

$$Y' = \begin{pmatrix} 1 & -3 \\ -2 & 2 \end{pmatrix} Y, \quad Y(0) = \begin{pmatrix} 0 \\ 5 \end{pmatrix}$$

Los valores propios son

$$\lambda_{1,2} = \frac{\text{Tr} \pm \sqrt{\text{Tr}^2 - 4 \det}}{2} = \frac{3 \pm \sqrt{9 + 16}}{2}$$

Luego, $\lambda_1 = 4$ y $\lambda_2 = 1$.

Para $\lambda_1 = 4$, tenemos que

$$\begin{pmatrix} -3 & -3 \\ -2 & -2 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Un vector propio asociado es

$$V_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Para $\lambda_2 = 1$,

$$\begin{pmatrix} 0 & -3 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Un vector propio asociado es

$$V_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

La solución general del sistema es

$$Y(t) = c_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{4t} + c_2 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^t$$

Ahora usamos la condición inicial $Y(0) = (0, 5)^\top$:

$$Y(0) = c_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 5 \end{pmatrix}$$

Esto produce el sistema

$$\begin{cases} c_1 + c_2 = 0 \\ -c_1 + 2c_2 = 5 \end{cases}$$

De la primera ecuación,

$$c_1 = -c_2$$

Sustituyendo en la segunda,

$$-(-c_2) + 2c_2 = 5$$

$$3c_2 = 5 \quad \Rightarrow \quad c_2 = \frac{5}{3}$$

Por lo tanto,

$$c_1 = -\frac{5}{3}$$

Finalmente,

$$Y(t) = -\frac{5}{3} \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{4t} + \frac{5}{3} \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^t$$

$$\boxed{Y' = \begin{pmatrix} 3 & 1 & -2 \\ -1 & 2 & 1 \\ 4 & 1 & -3 \end{pmatrix} Y, \quad Y(0) = \begin{pmatrix} 1 \\ 4 \\ 7 \end{pmatrix}}$$

Ecuación característica

$$\begin{aligned}
\det(A - \lambda I) &= \begin{vmatrix} 3 - \lambda & 1 & -2 \\ -1 & 2 - \lambda & 1 \\ 4 & 1 & -3 - \lambda \end{vmatrix} \\
&= (3 - \lambda) \begin{vmatrix} 2 - \lambda & 1 \\ 1 & -3 - \lambda \end{vmatrix} - 1 \begin{vmatrix} -1 & 1 \\ 4 & -3 - \lambda \end{vmatrix} + (-2) \begin{vmatrix} -1 & 2 - \lambda \\ 4 & 1 \end{vmatrix} \\
&= (3 - \lambda)[(2 - \lambda)(-3 - \lambda) - 1] - [(-1)(-3 - \lambda) - 4] + (-2)[(-1)(1) - 4(2 - \lambda)] \\
&= (3 - \lambda)(\lambda^2 + \lambda - 7) - (\lambda - 1) - 2(4\lambda - 9) \\
&= (3 - \lambda)(\lambda^2 + \lambda - 7) - \lambda + 1 - 8\lambda + 18 \\
&= (3 - \lambda)(\lambda^2 + \lambda - 7) - 9\lambda + 19 \\
&= (3\lambda^2 + 3\lambda - 21 - \lambda^3 - \lambda^2 + 7\lambda) - 9\lambda + 19 \\
&= -\lambda^3 + 2\lambda^2 + \lambda - 2 \\
&= -(\lambda^3 - 2\lambda^2 - \lambda + 2) \\
&= -(\lambda^2(\lambda - 2) - (\lambda - 2)) \\
&= -(\lambda - 2)(\lambda^2 - 1) \\
&= -(2 - \lambda)(\lambda - 1)(\lambda + 1)
\end{aligned}$$

Por lo tanto,

$$\lambda_1 = 2, \quad \lambda_2 = 1, \quad \lambda_3 = -1$$

Vectores propios

Para $\lambda_1 = 2$

$$(A - 2I) = \begin{pmatrix} 1 & 1 & -2 \\ -1 & 0 & 1 \\ 4 & 1 & -5 \end{pmatrix}$$

Resolvemos $(A - 2I)v = 0$:

$$\begin{cases} v_1 + v_2 - 2v_3 = 0 \\ -v_1 + v_3 = 0 \end{cases}$$

De la segunda ecuación,

$$v_1 = v_3$$

Sustituyendo en la primera,

$$v_3 + v_2 - 2v_3 = 0 \quad \Rightarrow \quad v_2 = v_3$$

Tomando $v_3 = 1$,

$$V_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Para $\lambda_2 = 1$

$$(A - I) = \begin{pmatrix} 2 & 1 & -2 \\ -1 & 1 & 1 \\ 4 & 1 & -4 \end{pmatrix}$$

Resolviendo $(A - I)v = 0$:

$$\begin{cases} 2v_1 + v_2 - 2v_3 = 0 \\ -v_1 + v_2 + v_3 = 0 \end{cases}$$

De la segunda ecuación,

$$v_2 = v_1 - v_3$$

Sustituyendo en la primera,

$$2v_1 + (v_1 - v_3) - 2v_3 = 0$$

$$3v_1 - 3v_3 = 0 \quad \Rightarrow \quad v_1 = v_3$$

Entonces

$$v_2 = 0$$

Tomando $v_3 = 1$,

$$V_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

Para $\lambda_3 = -1$

$$(A + I) = \begin{pmatrix} 4 & 1 & -2 \\ -1 & 3 & 1 \\ 4 & 1 & -2 \end{pmatrix}$$

Resolviendo $(A + I)v = 0$:

$$\begin{cases} 4v_1 + v_2 - 2v_3 = 0 \\ -v_1 + 3v_2 + v_3 = 0 \end{cases}$$

De la segunda ecuación,

$$v_1 = 3v_2 + v_3$$

Sustituyendo en la primera,

$$4(3v_2 + v_3) + v_2 - 2v_3 = 0$$

$$13v_2 + 2v_3 = 0 \quad \Rightarrow \quad v_3 = -\frac{13}{2}v_2$$

Tomando $v_2 = 2$,

$$V_3 = \begin{pmatrix} -7 \\ 2 \\ -13 \end{pmatrix}$$

Solución general

$$Y(t) = c_1 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} e^{2t} + c_2 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} e^t + c_3 \begin{pmatrix} -7 \\ 2 \\ -13 \end{pmatrix} e^{-t}$$

Condición inicial

$$Y(0) = c_1 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + c_3 \begin{pmatrix} -7 \\ 2 \\ -13 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \\ 7 \end{pmatrix}$$

Esto produce el sistema

$$\begin{cases} c_1 + c_2 - 7c_3 = 1 \\ c_1 + 2c_3 = 4 \\ c_1 + c_2 - 13c_3 = 7 \end{cases}$$

Restando la primera ecuación de la tercera,

$$-6c_3 = 6 \quad \Rightarrow \quad c_3 = -1$$

De la segunda ecuación,

$$c_1 + 2(-1) = 4 \quad \Rightarrow \quad c_1 = 6$$

Finalmente,

$$6 + c_2 + 7 = 1 \quad \Rightarrow \quad c_2 = -12$$

Solución del problema con valores iniciales

$$Y(t) = 6 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} e^{2t} - 12 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} e^t - \begin{pmatrix} -7 \\ 2 \\ -13 \end{pmatrix} e^{-t}$$

Ejercicio 5. Obtener la solución general de los siguientes sistemas lineales no homogéneos.

$$1. \quad Y' = \begin{pmatrix} -1 & 3 \\ -3 & 5 \end{pmatrix} Y + \begin{pmatrix} \cos(t) \\ \sin(t) \end{pmatrix} e^t \qquad 2. \quad Y' = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix} Y + \begin{pmatrix} e^t \\ e^{2t} \\ te^{3t} \end{pmatrix} e^t$$

Solución

$$Y' = \begin{pmatrix} -1 & 3 \\ -3 & 5 \end{pmatrix} Y + \begin{pmatrix} \cos t \\ \sin t \end{pmatrix} e^t$$

Sistema homogéneo

$$Y' = AY, \quad A = \begin{pmatrix} -1 & 3 \\ -3 & 5 \end{pmatrix}$$

Ecuación característica

$$\begin{aligned} \det(A - \lambda I) &= \begin{vmatrix} -1 - \lambda & 3 \\ -3 & 5 - \lambda \end{vmatrix} \\ &= (-1 - \lambda)(5 - \lambda) + 9 \\ &= \lambda^2 - 4\lambda + 4 \\ &= (\lambda - 2)^2 \end{aligned}$$

Por lo tanto,

$$\lambda = 2 \quad (\text{valor propio repetido})$$

Vector propio

Resolviendo $(A - 2I)V = 0$:

$$\begin{pmatrix} -3 & 3 \\ -3 & 3 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Se obtiene $v_1 = v_2$, luego el vector propio es:

$$V_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Vector propio generalizado

Buscamos V_2 resolviendo $(A - 2I)V_2 = V_1$:

$$\begin{pmatrix} -3 & 3 \\ -3 & 3 \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

La ecuación da $-3w_1 + 3w_2 = 1$, es decir $w_2 = w_1 + \frac{1}{3}$. Tomando $w_1 = 0$:

$$V_2 = \begin{pmatrix} 0 \\ 1 \\ \frac{1}{3} \end{pmatrix}$$

Soluciones del sistema homogéneo

$$Y_1(t) = e^{2t} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad Y_2(t) = e^{2t} \left(t \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \\ \frac{1}{3} \end{pmatrix} \right)$$

Matriz fundamental

$$\hat{Y}(t) = \begin{pmatrix} e^{2t} & te^{2t} \\ e^{2t} & \left(t + \frac{1}{3}\right)e^{2t} \end{pmatrix}$$

Variación de parámetros

El término no homogéneo es

$$G(t) = \begin{pmatrix} \cos t \\ \sin t \end{pmatrix} e^t$$

Usamos

$$Y_p(t) = \hat{Y}(t) \int \hat{Y}^{-1}(t) G(t) dt$$

Determinante de $\hat{Y}$:

$$\det \hat{Y} = e^{2t} \cdot \left(t + \frac{1}{3}\right) e^{2t} - te^{2t} \cdot e^{2t} = e^{4t} \left(t + \frac{1}{3} - t\right) = \frac{1}{3} e^{4t}$$

Adjunta de $\hat{Y}$:

$$\text{adj } \hat{Y} = \begin{pmatrix} \left(t + \frac{1}{3}\right) e^{2t} & -te^{2t} \\ -e^{2t} & e^{2t} \end{pmatrix}$$

Inversa:

$$\hat{Y}^{-1} = \frac{3}{e^{4t}} \begin{pmatrix} \left(t + \frac{1}{3}\right) e^{2t} & -te^{2t} \\ -e^{2t} & e^{2t} \end{pmatrix} = \frac{3}{e^{2t}} \begin{pmatrix} t + \frac{1}{3} & -t \\ -1 & 1 \end{pmatrix}$$

Producto $\hat{Y}^{-1}G$:

$$\begin{aligned}\hat{Y}^{-1}(t) G(t) &= \frac{3}{e^{2t}} \begin{pmatrix} t + \frac{1}{3} & -t \\ -1 & 1 \end{pmatrix} \begin{pmatrix} \cos t \\ \sin t \end{pmatrix} e^t \\ &= \frac{3}{e^t} \begin{pmatrix} (t + \frac{1}{3}) \cos t - t \sin t \\ -\cos t + \sin t \end{pmatrix} \\ &= 3e^{-t} \begin{pmatrix} t(\cos t - \sin t) + \frac{1}{3} \cos t \\ \sin t - \cos t \end{pmatrix}\end{aligned}$$

Integración (por partes donde sea necesario):

$$\begin{aligned}\int 3e^{-t}(\sin t - \cos t) dt &= 3 \int e^{-t}(\sin t - \cos t) dt \\ &= -3e^{-t} \sin t + C_2\end{aligned}$$

usando que $\int e^{-t}(\sin t - \cos t) dt = -e^{-t} \sin t + C$.

Para la primera componente:

$$\int 3e^{-t} [t(\cos t - \sin t) + \frac{1}{3} \cos t] dt = 3 \int t e^{-t}(\cos t - \sin t) dt + \int e^{-t} \cos t dt$$

Usando $\int e^{-t}(\cos t - \sin t) dt = e^{-t} \sin t$ y por partes:

$$3 \int t e^{-t}(\cos t - \sin t) dt = 3t e^{-t} \sin t - 3 \int e^{-t} \sin t dt = 3t e^{-t} \sin t + \frac{3}{2} e^{-t}(\sin t + \cos t)$$

y $\int e^{-t} \cos t dt = \frac{1}{2} e^{-t}(\sin t + \cos t)$, por lo que la primera componente es:

$$3t e^{-t} \sin t + 2e^{-t}(\sin t + \cos t) + C_1$$

Vector de coeficientes:

$$\mathbf{c}(t) = \begin{pmatrix} 3t e^{-t} \sin t + 2e^{-t}(\sin t + \cos t) \\ -3e^{-t} \sin t \end{pmatrix}$$

Solución particular $Y_p = \hat{Y} \mathbf{c}$:

$$\begin{aligned}Y_p(t) &= \begin{pmatrix} e^{2t} & t e^{2t} \\ e^{2t} & (t + \frac{1}{3}) e^{2t} \end{pmatrix} \begin{pmatrix} 3t e^{-t} \sin t + 2e^{-t}(\sin t + \cos t) \\ -3e^{-t} \sin t \end{pmatrix} \\ &= e^t \begin{pmatrix} 3t \sin t + 2(\sin t + \cos t) - 3t \sin t \\ 3t \sin t + 2(\sin t + \cos t) - 3(t + \frac{1}{3}) \sin t \end{pmatrix} \\ &= e^t \begin{pmatrix} 2 \sin t + 2 \cos t \\ 2 \cos t + \sin t \end{pmatrix}\end{aligned}$$

Solución general

$$Y(t) = c_1 e^{2t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 e^{2t} \left(t \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 0 \\ \frac{1}{3} \end{pmatrix} \right) + e^t \begin{pmatrix} 2 \cos t + 2 \sin t \\ 2 \cos t + \sin t \end{pmatrix}$$

$$Y' = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix} Y + \begin{pmatrix} e^t \\ e^{2t} \\ te^{3t} \end{pmatrix} e^t$$

Sistema homogéneo

$$Y' = AY, \quad A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

Ecuación característica

$$\begin{aligned} \det(A - \lambda I) &= \begin{vmatrix} 1 - \lambda & 1 & 0 \\ 1 & 1 - \lambda & 0 \\ 0 & 0 & 3 - \lambda \end{vmatrix} \\ &= (1 - \lambda)^2(3 - \lambda) - (3 - \lambda) \\ &= (3 - \lambda)[(1 - \lambda)^2 - 1] \\ &= (3 - \lambda)[\lambda^2 - 2\lambda + 1 - 1] \\ &= (3 - \lambda)(\lambda^2 - 2\lambda) \\ &= (3 - \lambda)(\lambda - 2)\lambda \end{aligned}$$

Por lo tanto los valores propios son:

$$\lambda_1 = 0, \quad \lambda_2 = 2, \quad \lambda_3 = 3$$

Vectores propios

Para $\lambda_1 = 0$: resolvemos $(A - 0 \cdot I)V = 0$:

$$\begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \mathbf{0} \implies \begin{cases} v_1 + v_2 = 0 \\ v_3 = 0 \end{cases} \implies V_1 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$$

Para $\lambda_2 = 2$: resolvemos $(A - 2I)V = 0$:

$$\begin{pmatrix} -1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \mathbf{0} \implies \begin{cases} -v_1 + v_2 = 0 \\ v_3 = 0 \end{cases} \implies V_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

Para $\lambda_3 = 3$: resolvemos $(A - 3I)V = 0$:

$$\begin{pmatrix} -2 & 1 & 0 \\ 1 & -2 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \mathbf{0} \implies \begin{cases} -2v_1 + v_2 = 0 \\ v_1 - 2v_2 = 0 \end{cases} \implies v_1 = v_2 = 0 \implies V_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Soluciones del sistema homogéneo

$$Y_1(t) = e^{0 \cdot t} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \quad Y_2(t) = e^{2t} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad Y_3(t) = e^{3t} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Matriz fundamental

$$\hat{Y}(t) = \begin{pmatrix} 1 & e^{2t} & 0 \\ -1 & e^{2t} & 0 \\ 0 & 0 & e^{3t} \end{pmatrix}$$

Variación de parámetros

El término no homogéneo es

$$G(t) = \begin{pmatrix} e^t \\ e^{2t} \\ te^{3t} \end{pmatrix} e^t = \begin{pmatrix} e^{2t} \\ e^{3t} \\ te^{4t} \end{pmatrix}$$

Usamos

$$Y_p(t) = \hat{Y}(t) \int \hat{Y}^{-1}(t) G(t) dt$$

Determinante de $\hat{Y}$ (expansión por cofactores de la tercera columna):

$$\det \hat{Y} = e^{5t} - e^{2t}(-e^{3t}) = 2e^{5t}$$

Matriz de cofactores de $\hat{Y}$:

Calculamos los nueve cofactores $A_{ij} = (-1)^{i+j} M_{ij}$:

$$\begin{aligned} A_{11} &= + \begin{vmatrix} e^{2t} & 0 \\ 0 & e^{3t} \end{vmatrix} = e^{5t}, & A_{12} &= - \begin{vmatrix} -1 & 0 \\ 0 & e^{3t} \end{vmatrix} = e^{3t}, & A_{13} &= + \begin{vmatrix} -1 & e^{2t} \\ 0 & 0 \end{vmatrix} = 0 \\ A_{21} &= - \begin{vmatrix} e^{2t} & 0 \\ 0 & e^{3t} \end{vmatrix} = -e^{5t}, & A_{22} &= + \begin{vmatrix} 1 & 0 \\ 0 & e^{3t} \end{vmatrix} = e^{3t}, & A_{23} &= - \begin{vmatrix} 1 & e^{2t} \\ 0 & 0 \end{vmatrix} = 0 \\ A_{31} &= + \begin{vmatrix} e^{2t} & 0 \\ e^{2t} & 0 \end{vmatrix} = 0, & A_{32} &= - \begin{vmatrix} 1 & 0 \\ -1 & 0 \end{vmatrix} = 0, & A_{33} &= + \begin{vmatrix} 1 & e^{2t} \\ -1 & e^{2t} \end{vmatrix} = 2e^{2t} \end{aligned}$$

Entonces, la matriz de cofactores es

$$\text{adj } \hat{Y} = \begin{pmatrix} e^{5t} & e^{3t} & 0 \\ -e^{5t} & e^{3t} & 0 \\ 0 & 0 & 2e^{2t} \end{pmatrix}$$

La adjunta es la transpuesta de la matriz de cofactores:

$$\text{adj } \hat{Y} = \begin{pmatrix} e^{5t} & -e^{5t} & 0 \\ e^{3t} & e^{3t} & 0 \\ 0 & 0 & 2e^{2t} \end{pmatrix}$$

Inversa:

$$\hat{Y}^{-1} = \frac{1}{2e^{5t}} \begin{pmatrix} e^{5t} & -e^{5t} & 0 \\ e^{3t} & e^{3t} & 0 \\ 0 & 0 & 2e^{2t} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} & 0 \\ \frac{1}{2}e^{-2t} & \frac{1}{2}e^{-2t} & 0 \\ 0 & 0 & e^{-3t} \end{pmatrix}$$

Producto $\hat{Y}^{-1}G$:

$$\begin{aligned} \hat{Y}^{-1}G &= \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} & 0 \\ \frac{1}{2}e^{-2t} & \frac{1}{2}e^{-2t} & 0 \\ 0 & 0 & e^{-3t} \end{pmatrix} \begin{pmatrix} e^{2t} \\ e^{3t} \\ te^{4t} \end{pmatrix} \\ &= \begin{pmatrix} \frac{1}{2}e^{2t} - \frac{1}{2}e^{3t} \\ \frac{1}{2}e^{-2t} \cdot e^{2t} + \frac{1}{2}e^{-2t} \cdot e^{3t} \\ e^{-3t} \cdot te^{4t} \end{pmatrix} = \begin{pmatrix} \frac{1}{2}(e^{2t} - e^{3t}) \\ \frac{1}{2}(1 + e^t) \\ te^t \end{pmatrix} \end{aligned}$$

Integración componente a componente:

$$u_1 = \int \frac{1}{2}(e^{2t} - e^{3t}) dt = \frac{1}{4}e^{2t} - \frac{1}{6}e^{3t}$$

$$u_2 = \int \frac{1}{2}(1 + e^t) dt = \frac{1}{2}t + \frac{1}{2}e^t$$

$$u_3 = \int te^t dt = (t-1)e^t$$

Solución particular:

$$\begin{aligned} Y_p(t) &= \begin{pmatrix} 1 & e^{2t} & 0 \\ -1 & e^{2t} & 0 \\ 0 & 0 & e^{3t} \end{pmatrix} \begin{pmatrix} \frac{1}{4}e^{2t} - \frac{1}{6}e^{3t} \\ \frac{1}{2}t + \frac{1}{2}e^t \\ (t-1)e^t \end{pmatrix} \\ &= \begin{pmatrix} \frac{1}{4}e^{2t} - \frac{1}{6}e^{3t} + \frac{1}{2}te^{2t} + \frac{1}{2}e^{3t} \\ -\frac{1}{4}e^{2t} + \frac{1}{6}e^{3t} + \frac{1}{2}te^{2t} + \frac{1}{2}e^{3t} \\ (t-1)e^{4t} \end{pmatrix} \\ &= \begin{pmatrix} \frac{1}{4}e^{2t} + \frac{1}{2}te^{2t} + \frac{1}{3}e^{3t} \\ -\frac{1}{4}e^{2t} + \frac{1}{2}te^{2t} + \frac{2}{3}e^{3t} \\ (t-1)e^{4t} \end{pmatrix} \end{aligned}$$

Solución general

$$Y(t) = c_1 \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + c_2 e^{2t} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + c_3 e^{3t} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} + \begin{pmatrix} \frac{1}{4}e^{2t} + \frac{1}{2}te^{2t} + \frac{1}{3}e^{3t} \\ -\frac{1}{4}e^{2t} + \frac{1}{2}te^{2t} + \frac{2}{3}e^{3t} \\ (t-1)e^{4t} \end{pmatrix}$$

Ejercicio 6. Traza y determinante en términos de los valores propios.

Sea $A \in \mathbb{C}^{n \times n}$ una matriz cuadrada y sean

$$\lambda_1, \lambda_2, \dots, \lambda_n$$

sus valores propios, contados con multiplicidad algebraica.

1. Demuestre que la traza de A puede expresarse como la suma de sus valores propios:

$$\text{Tr}(A) = \sum_{i=1}^n \lambda_i$$

2. Demuestre que el determinante de A puede expresarse como el producto de sus valores propios:

$$\det(A) = \prod_{i=1}^n \lambda_i$$

(*Hint:* Utilice la descomposición de Schur y el hecho de que, para matrices triangulares, el determinante es el producto de los elementos de su diagonal principal.)

Proof. Por la descomposición de Schur, toda matriz $A \in \mathbb{C}^{n \times n}$ es semejante a una matriz triangular superior. Es decir, existe una matriz invertible Q tal que

$$A = QTQ^{-1}$$

donde

$$T = \begin{pmatrix} \lambda_1 & t_{12} & \cdots & t_{1n} \\ 0 & \lambda_2 & \cdots & t_{2n} \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}$$

es triangular superior. Además, los elementos de la diagonal principal de T son los valores propios de A (contados con multiplicidad algebraica). Esto se debe a que, como A y T son matrices semejantes, comparten los mismos valores propios, y dado que en toda matriz triangular superior los valores propios coinciden con los elementos de su diagonal principal, entonces $\lambda_1, \dots, \lambda_n$ son precisamente los valores propios de A .

(1) Como A y T son matrices semejantes, tienen la misma traza. Entonces,

$$\text{Tr}(A) = \text{Tr}(T) = \sum_{i=1}^n \lambda_i$$

(2) De manera análoga, el determinante es invariante bajo semejanza:

$$\det(A) = \det(T)$$

Como T es triangular superior, su determinante es el producto de los elementos de la diagonal principal, por lo que

$$\det(T) = \prod_{i=1}^n \lambda_i$$

Por lo tanto,

$$\det(A) = \prod_{i=1}^n \lambda_i$$

□

Ejercicio 7. Polinomio característico y valores propios de una matriz 2×2 .

Sea $A \in \mathbb{R}^{2 \times 2}$ una matriz real.

1. Demuestre que el polinomio característico de A está dado por

$$p_A(\lambda) = \lambda^2 - \text{Tr}(A)\lambda + \det(A)$$

donde $\text{Tr}(A)$ denota la traza de A y $\det(A)$ su determinante.

2. A partir del resultado anterior, pruebe que los valores propios de A están dados por

$$\lambda_{1,2} = \frac{\text{Tr}(A) \pm \sqrt{\text{Tr}^2(A) - 4\det(A)}}{2}$$

Proof. Sea

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

(1) El polinomio característico es

$$\begin{aligned} p_A(\lambda) &= \begin{vmatrix} a - \lambda & b \\ c & d - \lambda \end{vmatrix} \\ &= (a - \lambda)(d - \lambda) - bc \\ &= \lambda^2 - (a + d)\lambda + (ad - bc) \\ &= \lambda^2 - \operatorname{Tr}(A)\lambda + \det(A) \end{aligned}$$

donde hemos usado que $a + d = \operatorname{Tr}(A)$ y $ad - bc = \det(A)$.

(2) Los valores propios de A son las raíces de $p_A(\lambda) = 0$, es decir:

$$\lambda^2 - \operatorname{Tr}(A)\lambda + \det(A) = 0$$

Aplicando la fórmula general:

$$\lambda_{1,2} = \frac{\operatorname{Tr}(A) \pm \sqrt{\operatorname{Tr}^2(A) - 4\det(A)}}{2}$$

□